Boundary Value Problems and Sturm–Liouville Theory

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Condensed Cheat Sheet: Two-Point BVPs, Eigenvalues $\lambda_n = (n\pi/L)^2$, and the Sturm–Liouville Operator $(p\,y')' + (q + \lambda w)y = 0$

Unit 17 at a Glance: From Side Conditions to a Functional Basis

An initial value problem pins a solution down with all its side data at one point; a boundary value problem spreads that data across two endpoints, and existence and uniqueness are no longer guaranteed. The interesting case is the homogeneous eigenvalue BVP $X'' + \lambda X = 0$, which is silent (only $X \equiv 0$) except at a discrete ladder of eigenvalues λ_n, where it rings with eigenfunctions X_n. Those eigenfunctions are mutually *orthogonal*, and orthogonality is the master key: it turns the infinite expansion $f = \sum c_n y_n$ into a list of one-line coefficient integrals. The Sturm–Liouville form $(p\,y')' + (q + \lambda w)y = 0$ packages every such problem and guarantees the eigenfunctions form a complete, weighted-orthogonal basis.

- **IVP vs. BVP** — moving the side conditions to two endpoints can make a linear problem have *no* solution, *one*, or *infinitely many*.
- **Eigenvalues / eigenfunctions** — $X'' + \lambda X = 0$, $X(0) = X(L) = 0$ is nontrivial only at $\lambda_n = (n\pi/L)^2$ with $X_n = \sin\frac{n\pi x}{L}$.
- **The shooting method** — recast a BVP as an IVP with a *guessed* initial slope, then adjust the slope until the far boundary condition is met.
- **Orthogonality** — $\int_a^b y_n y_m \, w \, dx = 0$ for $n \neq m$ makes each coefficient $c_n = \langle f, y_n\rangle / \langle y_n, y_n\rangle$ an independent integral.
- **Sturm–Liouville** — $(p\,y')' + (q + \lambda w)y = 0$ has real eigenvalues $\lambda_n \to \infty$ and w-orthogonal eigenfunctions that form a complete basis $f = \sum c_n y_n$.

Part I — Condensed Cheat Sheet

17.1 Initial versus Boundary Value Problems

Side conditions at one point (IVP) versus two endpoints (BVP); Dirichlet (y), Neumann (y'), and Robin conditions; the loss of the existence–uniqueness guarantee; steady-state heat conduction as the model BVP.

IVP versus BVP — Where the Data Lives

IVP: $y'' = F(x, y, y')$, $y(x_0) = y_0$, $y'(x_0) = y_0'$ (all data at one point x_0); **BVP:** $y(a) = \alpha$, $y(b) = \beta$ (d

The Three Standard Boundary Conditions

Dirichlet: $y(a) = \alpha$ (fixes the value); Neumann: $y'(a) = \alpha$ (fixes the slope/flux); Robin: $c_1 y(a) + c_2 y'(a) = \alpha$

Existence and Uniqueness — The Contrast

A linear IVP (continuous coefficients) $\Rightarrow$ **exactly one** solution; a linear BVP $\Rightarrow$ **none, one, or infinitely**

Method — Solve a Linear Two-Point BVP

1. Find the general solution of the ODE, carrying both arbitrary constants c_1, c_2.
2. Impose the condition at the left endpoint $x = a$ to get one equation in c_1, c_2.
3. Impose the condition at the right endpoint $x = b$ to get a second equation.
4. Solve the 2×2 system: a unique (c_1, c_2) gives one solution; an inconsistent system gives none; a dependent system gives infinitely many.

Key Takeaway

An initial value problem stacks all its side data at one point and is rewarded with a unique solution; a boundary value problem spreads the data across two endpoints and forfeits that guarantee. The same ODE can then have no solution, one, or infinitely many — which is exactly what makes BVPs the natural home for eigenvalues.

Common Pitfall

Do not carry the IVP intuition — "a second-order equation always has a unique solution given two conditions" — into BVPs. Two boundary conditions need not pin down a unique solution. Whether you get none, one, or infinitely many depends on how the conditions interact with the general solution, not merely on counting constants.

17.2 Two-Point Boundary Value Problems

The standard solve-then-impose method for linear two-point BVPs; the three possible outcomes; homogeneous BVPs and the omnipresent trivial solution; nonhomogeneous BVPs with forcing.

The Linear Two-Point BVP

$$y'' + P(x)y' + Q(x)y = R(x), \qquad y(a) = \alpha, \quad y(b) = \beta; \qquad y = c_1 y_1 + c_2 y_2 + y_p \text{ before imposing the BCs.}$$

Three Possible Outcomes

$$\det \begin{pmatrix} y_1(a) & y_2(a) \\ y_1(b) & y_2(b) \end{pmatrix} \neq 0 \;\Rightarrow\; \textbf{unique}; \qquad = 0 \;\Rightarrow\; \textbf{none or } \infty\text{-many (depending on } \alpha, \beta).$$

Homogeneous BVP — The Trivial Solution

$$y'' + \lambda y = 0, \;\; y(0) = y(L) = 0 \;\Rightarrow\; y \equiv 0 \text{ always solves it; nontrivial } y \text{ exist only at special } \lambda.$$

Method — Resolve a Two-Point BVP

1. Solve the ODE completely, including a particular solution y_p if the equation is nonhomogeneous.
2. Substitute the two boundary conditions to obtain a linear system in c_1, c_2.
3. Compute the boundary determinant: nonzero means a unique solution exists — solve for c_1, c_2.
4. If the determinant is zero, test consistency: an impossible equation gives no solution, a redundant one gives a one-parameter family of solutions.

Key Takeaway

Every linear two-point BVP is solved the same way: find the general solution first, then let the two boundary conditions decide the constants. The boundary determinant is the referee — nonzero awards a unique solution, while zero throws the problem into the no-solution or infinitely-many camp, the very fork that produces eigenvalues.

Common Pitfall

Never impose a boundary condition before you have the *full* general solution (including y_p). Dropping a homogeneous piece or the particular solution silently discards the freedom the BCs need, and you will report a unique answer where the true problem has none or infinitely many.

17.3 Eigenvalues and Eigenfunctions of Boundary Value Problems

The eigenvalue BVP $X''+\lambda X = 0$ with homogeneous BCs; the three-case analysis $\lambda < 0,\ \lambda = 0,\ \lambda > 0$; Dirichlet eigenvalues $\lambda_n = (n\pi/L)^2$ with eigenfunctions $\sin(n\pi x/L)$; the Neumann variant with cosines.

The Eigenvalue Boundary Value Problem

$$X''+\lambda X = 0, \qquad X(0) = 0,\ X(L) = 0; \qquad \lambda \text{ is an } \textit{eigenvalue} \text{ if a nontrivial } X \text{ (the } \textit{eigenfunction}\text{) exists.}$$

Dirichlet Spectrum

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \qquad X_n(x) = \sin\frac{n\pi x}{L}, \qquad n = 1, 2, 3, \ldots \quad (\lambda_n \to \infty).$$

Neumann Spectrum

$$X'(0) = X'(L) = 0: \quad \lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad X_n(x) = \cos\frac{n\pi x}{L}, \quad n = 0, 1, 2, \ldots\ (\lambda_0 = 0 \text{ allowed}, X_0 = 1).$$

Method — Find the Eigenvalues and Eigenfunctions

1. Split into three cases by the sign of λ, since the form of the general solution differs for $\lambda < 0$, $\lambda = 0$, and $\lambda > 0$.
2. For $\lambda < 0$ (write $\lambda = -\mu^2$) and $\lambda = 0$, impose the BCs and show only the trivial solution survives — no negative or zero eigenvalues here.
3. For $\lambda > 0$ (write $\lambda = \mu^2$) use $X = A\cos\mu x + B\sin\mu x$; the left BC kills A, and the right BC forces $\sin\mu L = 0$.
4. Solve the quantization condition $\mu L = n\pi$ to read off $\lambda_n = (n\pi/L)^2$ and the eigenfunctions $X_n = \sin(n\pi x/L)$.

Key Takeaway

A homogeneous BVP is a filter: it passes only the trivial solution except at a discrete ladder of eigenvalues $\lambda_n = (n\pi/L)^2$, where it resonates with the eigenfunctions $\sin(n\pi x/L)$. The negative and zero cases always collapse to $X \equiv 0$; the entire spectrum lives in $\lambda > 0$ and marches off to infinity.

Common Pitfall

You must check all three sign cases of λ — skipping the $\lambda \le 0$ analysis is only safe once you have *shown* it yields nothing. Also, $n = 0$ gives $X \equiv 0$ for Dirichlet (so start at $n = 1$), but $n = 0$ is a genuine eigenvalue $\lambda_0 = 0$ with $X_0 = 1$ in the Neumann case.

17.4 The Shooting Method

Recasting a BVP as an IVP with a guessed initial slope s; the boundary-mismatch function $F(s) = y(b; s) - \beta$; root-finding on F; for linear BVPs, exactly two shots plus linear interpolation; reuse of Runge–Kutta integrators.

BVP Recast as a Parameterized IVP

$$y'' = f(x, y, y'),\ y(a) = \alpha,\ y(b) = \beta \ \longrightarrow\ y'' = f,\ y(a) = \alpha,\ y'(a) = s \text{ (guess the slope } s\text{).}$$

The Boundary-Mismatch Function

$$F(s) = y(b;\, s) - \beta, \qquad \text{the shooting method seeks a root } F(s) = 0 \text{ so the far BC is met.}$$

Linear BVP — Two Shots and Interpolate

$$F \text{ affine in } s \Rightarrow s^\star = s_0 + (s_1 - s_0)\,\frac{\beta - y(b; s_0)}{y(b; s_1) - y(b; s_0)} \text{ (exact after two shots).}$$

Method — Shoot a Boundary Value Problem

1. Convert the BVP to an IVP by replacing the far condition $y(b) = \beta$ with a guessed slope $y'(a) = s$.
2. Integrate the IVP numerically (e.g. Runge–Kutta) from a to b and read off the overshoot/undershoot $F(s) = y(b; s) - \beta$.
3. Adjust s to drive $F(s) \to 0$: bisection or Newton in general; for a *linear* BVP, two shots fix the affine F exactly.
4. Interpolate to the exact slope $s^\star$ and integrate once more to produce the solution that meets both boundary conditions.

Key Takeaway

The shooting method trades a hard BVP for a familiar IVP by guessing the missing initial slope and then correcting it until the far boundary condition is hit. Root-finding on the mismatch $F(s) = y(b; s) - \beta$ does the correcting — and because F is affine for a linear BVP, two shots and a straight-line interpolation land it exactly, all while reusing standard Runge–Kutta machinery.

Common Pitfall

The clean two-shot interpolation is a linear-problem privilege. For a nonlinear BVP $F(s)$ is curved, so two shots do not suffice — you must iterate (Newton/bisection), and the IVP can be unstable, amplifying small slope errors over $[a, b]$ and ruining the far value.

17.5 Orthogonal Functions

The inner product $\langle f, g\rangle = \int_a^b fg\,dx$; orthogonality as a vanishing inner product; the sine orthogonality integral (0 for $m \neq n$, $L/2$ for $m = n$); the functional basis and the generalized Fourier coefficient $c_n = \langle f, y_n\rangle / \langle y_n, y_n\rangle$.

Inner Product and Orthogonality

$$\langle f, g\rangle = \int_a^b f(x)\,g(x)\,dx; \qquad f \perp g \iff \langle f, g\rangle = 0 \text{ (weighted: } \langle f, g\rangle_w = \int_a^b f\,g\,w\,dx).$$

The Sine Orthogonality Integral

$$\int_0^L \sin\frac{n\pi x}{L}\sin\frac{m\pi x}{L}\,dx = \begin{cases} 0, & m \neq n, \\ \dfrac{L}{2}, & m = n. \end{cases}$$

Functional Basis — Generalized Fourier Expansion

$$f(x) = \sum_{n=1}^{\infty} c_n\, y_n(x), \qquad c_n = \frac{\langle f,\, y_n\rangle_w}{\langle y_n,\, y_n\rangle_w} = \frac{\displaystyle\int_a^b f\, y_n\, w\, dx}{\displaystyle\int_a^b y_n^2\, w\, dx}.$$

Method — Expand a Function in an Eigenfunction Series

1. Identify the orthogonal eigenfunctions $\{y_n\}$ and the weight w for the underlying BVP (for a string, $y_n = \sin(n\pi x/L)$, $w = 1$).
2. Write the target expansion $f(x) = \sum_n c_n y_n(x)$ and isolate a single coefficient by taking the inner product of both sides with y_m.
3. Apply orthogonality: every term with $n \neq m$ vanishes, collapsing the infinite sum to one surviving term $c_m\langle y_m, y_m\rangle$.
4. Solve $c_m = \langle f, y_m\rangle/\langle y_m, y_m\rangle$ by evaluating the numerator and the (known) normalization integral in full.

Key Takeaway

Orthogonality is the engine of every eigenfunction expansion. Because $\langle y_n, y_m\rangle = 0$ for $n \neq m$, taking the inner product of $f = \sum c_m y_m$ with a single y_n annihilates the entire sum but one term, so each coefficient $c_n = \langle f, y_n\rangle/\langle y_n, y_n\rangle$ is computed *independently* by one integral — never by solving an infinite coupled system.

Common Pitfall

Do not forget to divide by the normalization $\langle y_n, y_n\rangle$ — here the $L/2$, not 1. Writing $c_n = \int_0^L f \sin(n\pi x/L)\, dx$ alone is off by the factor $2/L$. And when a weight $w(x) \neq 1$ is present, both integrals must carry it: $c_n = \int f y_n w\, dx / \int y_n^2 w\, dx$.

17.6 Sturm–Liouville Theory

The Sturm–Liouville form $(p\,y')' + (q + \lambda w)y = 0$; regular S–L problems and their guarantees (real eigenvalues, $\lambda_n \to \infty$, w-orthogonal eigenfunctions, completeness); the Lagrange-identity proof of orthogonality; converting an equation to S–L form.

The Sturm–Liouville Form

$$\big(p(x)\, y'\big)' + \big(q(x) + \lambda\, w(x)\big)y = 0, \qquad p, w > 0 \text{ on } [a, b], \quad \text{with homogeneous (separated) BCs.}$$

Guarantees of a Regular S–L Problem

(i) eigenvalues λ_n real, $\lambda_1 < \lambda_2 < \cdots \to \infty$; (ii) $\langle y_n, y_m \rangle_w = 0$ $(n \neq m)$; (iii) $\{y_n\}$ complete.

Putting an Equation in S–L Form

$$a_2 y'' + a_1 y' + a_0 y + \lambda y = 0 \xrightarrow{\mu = \frac{1}{a_2} e^{\int a_1/a_2\, dx}} (py')' + (q + \lambda w)y = 0, \quad p = \mu a_2,\ w = \mu.$$

Method — Prove Eigenfunction Orthogonality

1. Take two eigenpairs (λ_n, y_n) and (λ_m, y_m), $\lambda_n \neq \lambda_m$: $(py_n')' + (q + \lambda_n w)y_n = 0$ and $(py_m')' + (q + \lambda_m w)y_m = 0$.
2. Multiply the first by y_m and the second by y_n, then subtract; the q-terms cancel, leaving $(\lambda_n - \lambda_m) w\, y_n y_m = y_n (py_m')' - y_m (py_n')'$.
3. Recognize the right side as an exact derivative (Lagrange's identity): $y_n(py_m')' - y_m(py_n')' = \dfrac{d}{dx}\big[p\,(y_n y_m' - y_m y_n')\big]$.
4. Integrate over $[a, b]$: $(\lambda_n - \lambda_m) \displaystyle\int_a^b w\, y_n y_m\, dx = \big[p(y_n y_m' - y_m y_n')\big]_a^b$.
5. The separated homogeneous BCs make the bracket vanish at both ends; since $\lambda_n \neq \lambda_m$, conclude $\displaystyle\int_a^b y_n(x) y_m(x)\, w(x)\, dx = 0$.

Key Takeaway

Sturm–Liouville theory is the structural payoff of the unit: *any* second-order linear eigenvalue problem can be written as $(p\, y')' + (q + \lambda w)y = 0$, and once it is regular, three gifts follow automatically — the eigenvalues are real and unbounded, the eigenfunctions are orthogonal with respect to the weight w, and they form a complete basis. That completeness is exactly what licenses the generalized Fourier expansion $f = \sum c_n y_n$.

Common Pitfall

Eigenfunctions are orthogonal with respect to the weight $w(x)$, not plain dx — whenever $w \neq 1$, the inner product and every coefficient integral must carry it. Also, the orthogonality proof leans on the boundary bracket $[p(y_n y_m' - y_m y_n')]_a^b$ vanishing; this requires genuinely separated homogeneous BCs, so the guarantees can fail for a problem that is not a regular S–L problem.

Unit 17 Cheat-Sheet Takeaway

Move the side conditions to two endpoints and a unique solution is no longer promised; the homogeneous BVP $X'' + \lambda X = 0$ answers only at the eigenvalues $\lambda_n = (n\pi/L)^2$ with eigenfunctions $\sin(n\pi x/L)$. Those eigenfunctions are w-orthogonal, so $f = \sum c_n y_n$ with $c_n = \langle f, y_n\rangle / \langle y_n, y_n\rangle$ — and the Sturm–Liouville form $(p\,y')' + (q + \lambda w)y = 0$ guarantees all of it.